

2.2. Counting d.o.f

• $S=1$: A_μ s.d.h. $\delta A_\mu = \partial_\mu \varepsilon$, $\square A^\mu - \partial^\mu (\partial^\nu A_\nu) = 0$

$\rightarrow$ Lorentz gauge $\mathcal{D} \equiv \partial_\mu A^\mu = 0$

always reachable : $\delta \mathcal{D} = \square \varepsilon$ (solve $\square \varepsilon = -\mathcal{D}$ to get $\mathcal{D}=0$)

residual gauge transf : $\square \varepsilon = 0$

$\Rightarrow$ Maxwell becomes $\square A^\mu = 0$

$\rightarrow$ look at plane wave solutions : $A_\mu(x) = e_\mu(p) e^{ip_\mu x^\mu}$ pol. vector

$\square A_\mu = 0 \Rightarrow p^2 = 0$, $p_\mu = (p_+, 0, \dots, 0)$

Gauge transf : $\varepsilon(x) = \zeta(p) e^{ip \cdot x}$

$\rightarrow \delta e_\mu = ip_\mu \zeta$ s.d.h. $\neq 0$: $\delta e_+ = ip_+ \zeta$

$\Rightarrow$ choose $\zeta = \frac{ie_+}{p_+}$, $e_+ = 0$ (completely fixes gauge)

Lorentz condition :

$\rightarrow p^\mu e_\mu = 0$

$\Rightarrow p_+ p_- + p_- p_+ + e_i p_i \Rightarrow e_- = 0$ reps of little group $SO(D-2)$

$\Rightarrow$ only $D-2$ comp. remain : e_i , $i = 1, \dots, D-2$

• Arbitrariness 3

→ generalised de Donder gauge :

$$\mathcal{D}_{\mu_2 \dots \mu_s} \equiv \partial^\nu \phi_{\nu \mu_2 \dots \mu_s} - \frac{(s-1)}{2} \partial_{(\mu_2} \phi_{\mu_3 \dots \mu_s)} \nu^\nu = 0$$

→ gauge variation : $\delta \mathcal{D}_{\mu_2 \dots \mu_s} = \square \xi_{\mu_2 \dots \mu_s}$

⇒ set $\mathcal{D} = 0$ by solving wave equation iff $\square \xi = 0$
($\square \xi = 0$ because $\bar{\phi} = 0$)

→ residual gauge ξ satisfy $\square \xi_{\mu_2 \dots \mu_s} = 0$

→ $\mathcal{F} \dots$ equation becomes $\square \phi_{\mu_1 \dots \mu_s} - \underbrace{s \partial_{(\mu_1} \mathcal{D}_{\mu_2 \dots \mu_s)}}_{=0} = 0$

→ plane waves $\phi_{\mu_1 \dots \mu_s} = e_{\mu_1 \dots \mu_s} e^{ip \cdot x}$

$$\text{s.t.} \quad \delta e_{\mu_1 \dots \mu_s} = i p_{(\mu_1} \xi_{\mu_2 \dots \mu_s)}$$

⇒ can set all $e_{+\mu_2 \dots \mu_s} = 0$ (completely fixes the gauge)

→ de Donder $p_+ e_{-\mu_2 \dots \mu_s} - \left(\frac{s-1}{2}\right) p_{(\mu_2} e_{\mu_3 \dots \mu_s)} \nu^\nu = 0$

→ take no "+" in $\mu_2 \dots \mu_s$: $p_+ e_{-\mu_2 \dots \mu_s} = 0$

$$\Rightarrow \text{all } e_{-\mu_2 \dots \mu_s} = 0$$

→ take one "+" in $\mu_2 \dots \mu_s$: $p_+ e_{-\mu_3 \dots \mu_s} \nu^\nu = 0$

$$\Rightarrow \text{for } e = 0$$

2.3. Action principle

$\rightarrow$ want $\mathcal{S}[\phi]$ s.t. $\frac{\delta \mathcal{S}}{\delta \phi_{\mu_1 \dots \mu_s}} = 0 \iff \mathcal{F}_{\mu_1 \dots \mu_s} = 0$

and gauge invar. $\delta_{\xi} \mathcal{S} = 0$

Remark: $s=2$, $R_{\mu\nu} = 0$ doesn't follow from $\mathcal{S}$

$\mathcal{E}_{\mu\nu} \equiv R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 0$ follows from $\mathcal{S}$

$\Rightarrow R = 0$ (trace trick)

Remark: $0 = \delta_{\xi} \mathcal{S} = \int \frac{\delta \mathcal{S}}{\delta \phi_{\mu_1 \dots \mu_s}} \delta_{\xi} \phi_{\mu_1 \dots \mu_s}$

$= \int \frac{\delta \mathcal{S}}{\delta \phi_{\mu_1 \dots \mu_s}} \cdot s \partial_{\mu_1} \xi_{\mu_2 \dots \mu_s}$

$= (-s) \int \xi_{\mu_2 \dots \mu_s} \left(\partial_{\mu_1} \frac{\delta \mathcal{S}}{\delta \phi_{\mu_1 \dots \mu_s}} \right) \quad \forall \xi_{\mu_1 \dots \mu_s} \text{ symm. traceless}$

$\Rightarrow$ Noether identity: $\partial_{\mu_1} \frac{\delta \mathcal{S}}{\delta \phi_{\mu_1 \dots \mu_s}} \Big|_{\text{traceless part}} = 0$

Remark: this is a general feature of an action: gauge invariance is equivalent to e.o.m.s

(Einstein equations $\nabla^{\mu} \mathcal{E}_{\mu\nu} = 0$)

Solution: define the Einstein tensor

$\rightarrow \mathcal{G}_{\mu_1 \dots \mu_s} = \mathcal{F}_{\mu_1 \dots \mu_s} - \frac{s(s-1)}{4} \eta_{(\mu_1 \mu_2} \mathcal{F}_{\mu_3 \dots \mu_s)} \eta^{\nu}$

$$\Rightarrow S[\phi] = \frac{1}{2} \int d^D x \phi_{\mu_1 \dots \mu_s} \mathcal{G}^{\mu_1 \dots \mu_s}$$

$$\rightarrow \delta S = \frac{1}{2} \int (\phi \mathcal{G}[\delta\phi] + \delta\phi \mathcal{G}[\phi]) = \delta\phi \mathcal{G}[\phi]$$

$$\rightarrow \mathcal{G}_{\mu_1 \dots \mu_s} \text{ is doubly traceless} : \frac{\delta S}{\delta \phi^{\mu_1 \dots \mu_s}} = \mathcal{G}_{\mu_1 \dots \mu_s}$$

$$\rightarrow \mathcal{G} = 0 \Leftrightarrow \mathcal{F} = 0$$

$$\rightarrow \mathcal{F} \text{ satisfies } \partial^\nu \mathcal{F}_{\nu \mu_2 \dots \mu_s} - \frac{s-1}{2} \partial_{(\mu_2} \mathcal{F}_{\mu_3 \dots \mu_s)} \nu^\nu = 0$$

$$\Rightarrow \partial^\nu \mathcal{F}_{\nu \mu_2 \dots \mu_s} = \mathcal{U}_{(\mu_2 \mu_3} \mathcal{T}_{\mu_4 \dots \mu_s)}$$

$\Leftrightarrow$ gauge invariant action